

Orbital edge computing with onboard AI for bandwidth-efficient Earth observation in defense and aerospace.

Aerospace and defense industry × Remote and autonomous operations × Edge computing · OS122 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
76 is the world moving	50 can we play, can we win	HIGH 1 named in the evidence	€34.7m modelled evidence · per year	NEXT pilots published no volume pr...	L0 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Aerospace and defense industry, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is about deploying edge AI on low-Earth-orbit satellites to process Earth observation data in orbit, reducing the downlink bottleneck that currently delays or loses valuable imagery. It targets defense and aerospace customers who need faster access to intelligence from space. The time is right because recent research demonstrates that onboard vision-language models and collaborative orbital edge computing can summarize data before download, making bandwidth-efficient Earth observation feasible.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€42.1m €8.5m – €182m	€34.7m €7.0m – €151m	€521k €106k – €2.3m	modelled
Observed: tendered contract value, annualised	€45.7m	€45.7m	€686k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Aerospace and defense industry (EU27_2020)	3,354 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Edge computing	16.9 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	780 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

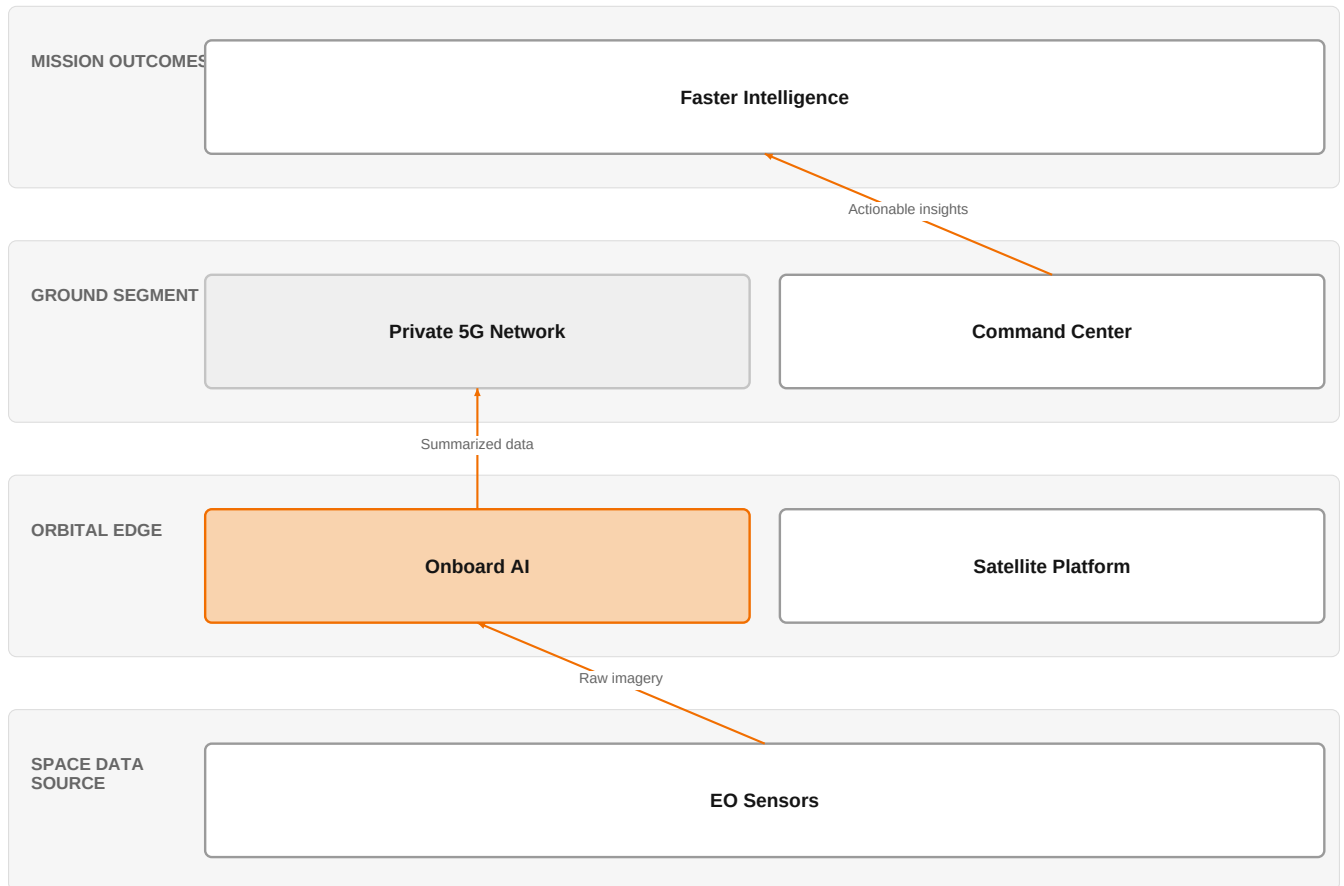
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€45.7m per year	observed	TED (Tenders Electronic Daily) · ted	2026-07-07..2026-07-08
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 3 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Orbital Edge AI for Earth Observation



This diagram shows how onboard AI on satellites reduces downlink load, with Orange providing secure ground connectivity.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Earth observation satellites now carry sensors that produce vast volumes of high-resolution data, but downlink capacity remains a critical bottleneck, causing latency and data loss. Recent research shows that onboard edge computing with AI can process data in orbit, providing edge intelligence close to space data sources. Studies propose 'Summarize First, Download Later' using onboard vision-language models, and collaborative orbital edge computing for energy-efficient processing. These advances are moving from theory to practical architectures, making orbital edge computing a credible option for defense and aerospace customers.

Sources: SIG-F5745ED14858 [arxiv.org 2026-08-07](https://arxiv.org/abs/2026.08.07); SIG-DEDC91C97937 [arxiv.org 2026-08-01](https://arxiv.org/abs/2026.08.01); SIG-45607AAD5C54 [arxiv.org 2026-07-28](https://arxiv.org/abs/2026.07.28)

Who buys, and why now

The buyer is a defense or aerospace program manager responsible for satellite-based intelligence, surveillance, and reconnaissance (ISR). The pain is operational: downlink bottlenecks delay critical imagery, and limited contact windows cause data loss, meaning decision-makers wait too long for actionable information. The trigger is a mission where time-sensitive data was lost or delayed, or a new satellite program where bandwidth requirements exceed downlink capacity.

Sources: SIG-F5745ED14858 [arxiv.org 2026-08-07](https://arxiv.org/abs/2026.08.07); SIG-45607AAD5C54 [arxiv.org 2026-07-28](https://arxiv.org/abs/2026.07.28)

Why it is hot now — every claim bound to a dated source

- LEO satellites with onboard edge computing can process data in orbit, reducing downlink bottlenecks. [SIG-DEDC91C97937, SIG-F5745ED14858]
- Modern Earth observation satellites produce vast data volumes but downlink capacity is a critical bottleneck, causing latency and data loss. [SIG-F5745ED14858]
- Orbital edge computing adds computational intelligence to LEO satellites to process data on orbit, providing edge intelligence close to space data sources. [SIG-DEDC91C97937]
- Earth observation satellites have limited downlink bandwidth that delays perceived revisiting cycles. [SIG-45607AAD5C54]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution combining its IoT and defense/security expertise with private mobile networks for ground segment connectivity. The engagement would start with a feasibility study using Orange Group researchers to assess the customer's satellite data pipeline and identify where onboard AI could reduce downlink load. Then, Orange would design and pilot an edge AI stack on the customer's satellite or a testbed, leveraging partnerships with NVIDIA, Dell, and HPE for compute hardware and AI software. Finally, Orange would integrate the solution with the customer's ground segment, using private mobile networks for secure data relay.

Why Orange can win

Orange brings unique strengths: deep defense and security expertise, a strong research arm, and experience with private mobile networks for secure, low-latency connectivity. The partnership with NVIDIA, Dell, and HPE provides access to leading edge AI hardware and software. However, the assets are thin in the sense that Orange does not have a proven orbital edge computing product; this would be a new build. The win would come from combining these assets into a trusted, sovereign solution for defense customers.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.3; 1 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competition is intense. AWS, a hyperscaler and Orange partner, offers edge computing with strong cloud integration. Deutsche Telekom/T-Systems and Lumen Technologies, telecom peers, have edge offerings. NTT DATA and Atos/Eviden, global systems integrators, are active in aerospace and defense. HPE, also a partner, sells edge computing hardware. Nokia and Schneider Electric are also in the edge space. The customer will compare on technical depth, security, and ability to deliver a sovereign solution.

Sources: SIG-701F806B6A7A International Journal 2025-07-30

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Edge computing; competes in OX: Smart Industries — also an Orange partner	evidenced	SIG-701F806B6A7A
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; competes in OX: Smart Industries	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing	structural	—

NTT DATA	Global systems integrator	sells Edge computing; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing; competes in OX: Smart Industries	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Edge computing; competes in OX: Smart Industries	structural	—
Atos / Eviden	Global systems integrator	active in Aerospace and defense industry	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is the current downlink capacity of your Earth observation satellites, and how much data is lost or delayed due to bandwidth limits?
- 2 Have you already evaluated onboard processing or AI for your satellites? If so, what were the results?
- 3 What is the typical latency from image capture to delivery for your most time-sensitive missions?
- 4 Are you planning any new satellite launches in the next 24 months where edge computing could be integrated?
- 5 What are your security and sovereignty requirements for data processing in space?
- 6 Who in your organization owns the satellite data pipeline and would be the champion for this initiative?

Objections you will meet

They will say	You can answer
Onboard AI is too risky for critical defense missions; we can't afford failures in space.	That's a valid concern. We would start with a pilot on a non-critical satellite or a ground testbed to prove reliability. Our approach is incremental, and we leverage proven hardware from partners like NVIDIA and Dell to minimize risk.
We already have a cloud-based processing pipeline that works; why change?	Your current pipeline works, but it is limited by downlink bandwidth. Onboard processing can reduce the data you need to download, speeding up delivery and reducing costs. We can show you a proof-of-concept using your own data to quantify the benefit.
This sounds expensive and unproven. What's the ROI?	The investment is significant, but the payoff is faster access to critical intelligence and less data loss. We can model the ROI based on your specific downlink constraints and mission value. Also, as the technology matures, costs will come down.

Risks and unknowns

The main risk is that orbital edge computing is still emerging; there is no proven large-scale deployment. The customer may be skeptical about the reliability of AI in space and the power constraints of satellites. Unknowns include the actual performance of onboard AI models in orbit, the cost of integrating edge hardware into satellites, and whether the customer's downlink bottleneck is severe enough to justify the investment. Also, regulatory and export control issues for defense applications could stall the project.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a defense or aerospace tender on remote autonomous operations, positioning Orange's private mobile networks and IoT expertise alongside NVIDIA, Dell, or HPE to deliver orbital edge computing that reduces downlink latency.
Sales	In a conversation with a defense or aerospace prime's satellite programs director, say: 'We're exploring how Orange's private mobile networks and edge computing expertise, combined with partners like NVIDIA, Dell, or HPE, could help you process Earth observation data in orbit to reduce downlink bottlenecks.'
Strategist / Innovator	Commission a deep-dive brief on orbital edge computing that maps Orange's IoT, defense and security expertise against the downlink bottleneck problem, and identifies which partner (NVIDIA, Dell, or HPE) offers the most credible path to a pilot with a defense or aerospace customer.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-E23019E5F6EE	2026-07-01	2026 6th International Co...	1	A Latency-Efficient NF Caching and Offloading Strategy for LEO Satellite Edge Computing
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-3772E07403F3	2026-01-02	openalex.org	1	Cloudburst AI: Orchestrating SRE-Driven AI Inference Across Multi-Cloud and Edge Zones
SIG-04BD044A1960	2025-11-18	openalex.org	1	Cognitive Edge Computing: A Comprehensive Survey on Optimizing Large Models and AI Agents for P...
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-D031822F4DA6	2025-09-09	openalex.org	1	Closed-Loop Network Control for Industrial Edge Computing: A Telemetry-Driven and AI-Based Appr...
SIG-701F806B6A7A	2025-07-30	International Journal of ...	1	Edge Computing in Networks: Reducing Latency Using AI-Driven Edge Computing Strategies
SIG-28B123DA3970	2025-06-30	Computing and Application...	1	EDGE COMPUTING AND CLOUD SYNERGY: OPTIMIZING LATENCY AND BANDWIDTH FOR IOT APPLICATIONS THROUGH...
SIG-37D6B973F20E	2026-08-12	bing.com	2	Lockheed Martin demonstrates AI-powered drone detection system using 5G networks
SIG-86EDC2F80FE3	2026-05-07	PRWeb	2	VIRTOWER Enhances Airspace Safety With Remote ID Drone Detection Upgrade - PRWeb
SIG-F5745ED14858	2026-08-07	arxiv.org	3	Summarize First, Download Later: Onboard VLMs for Bandwidth-Efficient Earth Observation
SIG-DEDC91C97937	2026-08-01	arxiv.org	3	Collaborative Orbital Edge Intelligence: A Decentralized Paradigm for Energy-Efficient Computin...
SIG-45607AAD5C54	2026-07-28	arxiv.org	3	Revisiting-Aware In-Orbit Edge Computing for Earth Observation
SIG-D29A41589B0F	2026-07-25	arxiv.org	3	Movable-Antenna Assisted Energy Minimization in UAV-Enabled Mobile Edge Computing Systems
SIG-575FE627E0DA	2026-07-11	arxiv.org	3	CSI-Assisted Edge SLAM Testbed Platform for 5G Connected Unmanned Autonomous Vehicles

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:12+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.